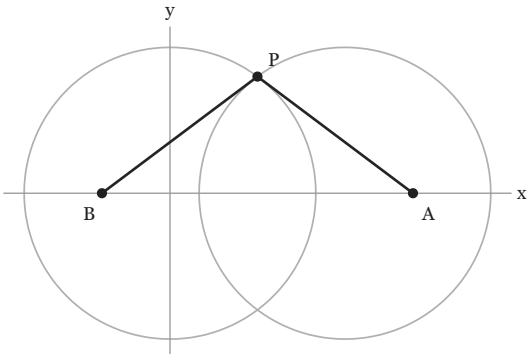
2026 Predictive Paper · Paper 1

Question 1

The circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$ intersect at P, which lies above the x-axis.

The tangents to the two circles at P meet the x-axis at A and B.

What is the area of triangle PAB?



The two tangents meet at P. Use the equations to establish exact coordinates.

A $\frac{32}{3}$

B 16

C $\frac{64}{3}$

D $\frac{100}{3}$

E 25

Worked solution · C

Subtracting the circle equations gives $x = 3$, so $P = (3, 4)$. A tangent is perpendicular to the radius. The two tangent equations are $3x + 4y = 25$ and $-3x + 4y = 7$.

Their x-intercepts are $25/3$ and $-7/3$. Thus $AB = 32/3$ and the perpendicular height from P is 4. The area is $\frac{1}{2} \times \frac{32}{3} \times 4 = \frac{64}{3}$.

Check: the gradients of the radii are $4/3$ and $-4/3$; those of the tangents are $-3/4$ and $3/4$. Both tangents pass through $(3, 4)$.

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Question 2

For which real values of a does the equation

$\sqrt{x+a} = x - 1$

have exactly one real solution, with that solution strictly between 1 and 3?

A $-\frac{5}{4} < a \leq -1$

B $-1 < a < 1$

C $a = -\frac{5}{4}$ or $-1 < a < 1$

D $-\frac{5}{4} \leq a < 1$

E $a = -\frac{5}{4}$ or $-1 \leq a \leq 1$

Worked solution · C

The original equation requires $x \geq 1$. Squaring gives $x^2 - 3x + 1 - a = 0$, with roots $x = (3 \pm \sqrt{5+4a})/2$. Every root of this quadratic with $x \geq 1$ satisfies the original equation, since then $x + a = (x-1)^2$.

At $a = -5/4$ there is one distinct root, $x = 3/2$. If $-5/4 < a \leq -1$, both roots are admissible, including $x = 1$ at $a = -1$. If $a > -1$, only the larger root is admissible. It is less than 3 exactly when $a < 1$.

Hence the answer is the isolated value $a = -5/4$, together with $-1 < a < 1$. The endpoints $a = -1$ and $a = 1$ fail for different reasons: two solutions, and a solution equal to 3, respectively.

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Question 3

A sequence is defined by $u_1 = 3$ and

$u_{n+1} = \frac{3u_n - 1}{u_n + 1}$ for every positive integer n.

What is $u_1 u_2 \cdots u_{20}$?

A 21

B 420

C 462

D 210

E 231

Worked solution · E

The first terms are $3, 2, 5/3, 3/2$. They suggest $u_n = (n+2)/n$. Substitution verifies this: $(3(n+2) - n)/((n+2) + n) = (n+3)/(n+1)$, and the initial term is correct. All denominators are positive.

Therefore the product is $\frac{3}{1}\frac{4}{2}\frac{5}{3}\cdots\frac{22}{20} = \frac{21 \times 22}{2} = 231$.

A second route is to put $v_n = 1/(u_n - 1)$. Then $v_{n+1} = v_n + 1/2$, so $v_n = n/2$ and the same formula follows. The recurrence preserves $u_n > 1$, justifying this substitution.

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Question 4

For $0 \le x \le 2\pi$, consider the equation

$$\frac{\sin^2 x}{1 - \cos x} = 2 \sin x$$

What is the sum of $\cos x$ over all its distinct real solutions in this interval?

A $-\frac{2}{5}$

B $\frac{8}{5}$

C $\frac{3}{5}$

D 0

E -1

Worked solution · A

The denominator excludes $x = 0$ and $x = 2\pi$. At every other point the equation is $1 + \cos x = 2 \sin x$. Write $s = \sin x$, so that $\cos x = 2s - 1$.

Using $s^2 + (2s - 1)^2 = 1$ gives $s(5s - 4) = 0$. The case $s = 0$ requires $\cos x = -1$, hence $x = \pi$. The other case has $s = 4/5$ and $\cos x = 3/5$, giving exactly one point in the first quadrant. The required sum is $-1 + 3/5 = -2/5$.

Both candidate points satisfy the original equation. Dividing by $\sin x$ would lose the solution $x = \pi$.

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Question 5

A continuous function f, defined for all real x, satisfies

$f(x) - f(-x) = 2x^3 - 6x$

What is $\int_{-2}^{2} x f(x)\,dx$?

A $\frac{16}{5}$

B $\frac{32}{5}$

C $-\frac{32}{5}$

D $-\frac{16}{5}$

E 0

Worked solution · D

Pair the contributions at x and $-x$. The integral becomes $\int_0^2 x[f(x) - f(-x)]\,dx = \int_0^2 (2x^4 - 6x^2)\,dx$.

This is $[2x^5/5 - 2x^3]_0^2 = 64/5 - 16 = -16/5$.

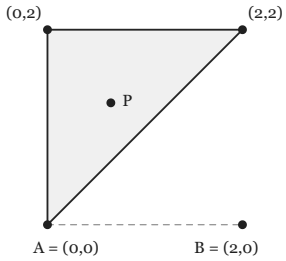
Equivalently, $f(x) = x^3 - 3x + g(x)$ for an even function g. The function $xg(x)$ is odd and contributes zero on the symmetric interval. The condition is feasible, for example with $g = 0$.

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Question 6

Let $A = (0, 0)$ and $B = (2, 0)$. A point P is chosen uniformly by area from the interior of the triangle with vertices $(0, 0)$, $(0, 2)$ and $(2, 2)$.

What is the probability that angle APB is greater than 90°?



P is chosen uniformly by area from the shaded triangle.

A $\frac{\pi-2}{4}$

B $\frac{4-\pi}{8}$

C $\frac{\pi}{8}$

D $\frac{\pi-2}{8}$

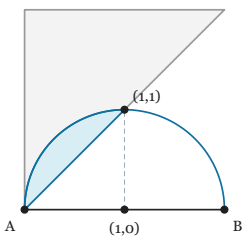
E $\frac{\pi}{4}$

Worked solution · D

The angle is obtuse exactly when P lies inside the circle with diameter AB, whose equation is $(x-1)^2 + y^2 = 1$. This also follows from the cosine rule: $AP^2 + BP^2 - AB^2 = 2(x^2 - 2x + y^2)$, which is negative precisely for an obtuse angle at P.

Within the sample triangle, the required region lies above $y = x$ and below the upper circular arc, from $(0, 0)$ to $(1, 1)$. It is a quarter-disc of radius 1 with a right triangle of area $1/2$ removed. Its area is $\pi/4 - 1/2$.

The sample triangle has area 2. Therefore the probability is $(\pi/4 - 1/2)/2 = (\pi - 2)/8$. Boundary points have area zero and do not affect the result.



The blue region has area π/4 − 1/2. The whole sample triangle has area 2.

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Question 7

The positive integer n is such that the coefficient of x^4 in

$(1-x)^n(1+x)^{n+2}$

is 27. What is the coefficient of x^6 in the same expression?

A -48

B 120

C -84

D -120

E 48

Worked solution · A

Rewrite the expression as $(1-x^2)^n(1+x)^2$. The coefficient of x^4 is $\binom{n}{2} - n = n(n-3)/2$. Thus $n^2 - 3n - 54 = 0$, so $n = 9$ since n is positive.

The coefficient of x^6 is $-\binom{9}{3} + \binom{9}{2} = -84 + 36 = -48$. The term $2x$ cannot contribute to an even power because $(1-x^2)^9$ contains only even powers.

Check: direct expansion for $n = 9$ gives coefficients 27 at x^4 and -48 at x^6. The other algebraic value, $n = -6$, is outside the question.

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Question 8

What is the total length of the locus of points (x, y) satisfying

$\log_2(5-y) + \log_2(y+1) = 2\log_2(|x|-2)$

A 9π

B 12π

C 3π

D 6π

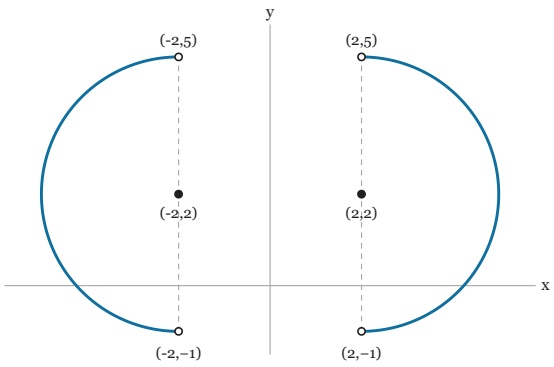
E 4π

Worked solution · D

The logarithms require $-1 < y < 5$ and $|x| > 2$. Combining them and exponentiating gives $(5-y)(y+1) = (|x|-2)^2$, or $(|x|-2)^2 + (y-2)^2 = 9$.

For $x > 2$, this is the right-hand semicircle of the circle centred at $(2, 2)$ with radius 3. For $x < -2$, it is the left-hand semicircle of the circle centred at $(-2, 2)$ with radius 3.

Each arc has length 3π. The four excluded endpoints have zero length. The total is 6π. The other semicircles violate $|x| > 2$.



Only |x| > 2 is allowed. Open circles show the excluded endpoints; omitting points does not change arc length.

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Question 9

In triangle ABC, the vertices are $A = (0, 0)$, $B = (6, 0)$ and $C = (0, 8)$.

At time $t = 0$, P starts at A and moves along AB at 3 units per second. At the same time, Q starts at B and moves along BC at 5 units per second. Each stops on reaching the end of its segment.

What is the least distance between P and Q during $0 \le t \le 2$?

A 4

B 3

C $\frac{6\sqrt{13}}{13}$

D $\frac{12}{5}$

E $\frac{12\sqrt{13}}{13}$

Worked solution · E

The length of BC is 10, so both journeys last 2 seconds. At time t, $P = (3t, 0)$ and $Q = (6 - 3t, 4t)$.

Their squared distance is $(6 - 6t)^2 + (4t)^2 = 52t^2 - 72t + 36 = 52(t - 9/13)^2 + 144/13$.

The minimum occurs at $t = 9/13$, which is inside the allowed interval. Thus the least distance is $12/\sqrt{13} = 12\sqrt{13}/13$. Checking the endpoints gives distances 6 and 10, both larger.

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Question 10

A histogram represents exactly 20 measurements. The class intervals and frequency densities are:

Class interval	$[0, 1)$	$[1, 3)$	$[3, 6]$
Frequency density	6	4	2

The mean of all 20 measurements is exactly 1.8. What is the greatest possible number of measurements that are at least 5?

A 3

B 5

C 8

D 4

E 6

Worked solution · B

Multiplying each density by its class width gives frequencies $6, 8, 6$. The sum of the measurements is $20(1.8) = 36$.

If k measurements are at least 5, they must be in the final class. The sum is at least $6(0) + 8(1) + (6 - k)3 + 5k = 26 + 2k$. Hence $36 \geq 26 + 2k$, so $k \leq 5$.

The bound is attainable: take six 0s, eight 1s, one 3 and five 5s. These have the stated frequencies and sum 36. Thus the greatest possible number is 5.

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Question 11

Two distinct tangents to the parabola $y = x^2$ pass through the point $(3, -3)$.

What is the area of the triangle formed by these two tangents and the y-axis?

A $36\sqrt{3}$

B $24\sqrt{3}$

C 54

D $18\sqrt{3}$

E $72\sqrt{3}$

Worked solution · A

The tangent at (t, t^2) is $y = 2tx - t^2$. Passing through $(3, -3)$ requires $t^2 - 6t - 3 = 0$, giving $t = 3 \pm 2\sqrt{3}$.

The two intercepts on the y-axis are the negatives of the squares of these values. Their separation is $(4\sqrt{3})(6) = 24\sqrt{3}$. The perpendicular distance of $(3, -3)$ from the y-axis is 3.

Therefore the area is $\frac{1}{2}(24\sqrt{3})(3) = 36\sqrt{3}$. Both contact parameters are real and different, so the specified triangle is nondegenerate.

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Question 12

Let $f(x) = |x^2 - 4x + 3|$. For which positive real values of k does

$f(f(x)) = k$

have exactly eight distinct real solutions?

A $0 < k < 1$

B $k > 3$

C $k = 1$

D $k = 3$

E $1 < k < 3$

Worked solution · C

First count solutions of $f(x) = y$ for $y \geq 0$. There are 2 for $y = 0$, 4 for $0 < y < 1$, 3 for $y = 1$, and 2 for $y > 1$. These follow from $x = 2 \pm \sqrt{1 + y}$ and, when $y \leq 1$, $x = 2 \pm \sqrt{1 - y}$, removing repeated roots.

Next solve $f(y) = k$ with $y \geq 0$, since $y = f(x)$. For $0 < k < 1$, the admissible values of y give $4 + 2 + 2 + 2 = 10$ roots. At $k = 1$, they are $2 - \sqrt{2}, 2, 2 + \sqrt{2}$, giving $4 + 2 + 2 = 8$ roots.

For $1 < k < 3$ the total is $4 + 2 = 6$; at $k = 3$ it is $2 + 2 = 4$; and for $k > 3$ it is 2. Distinct intermediate values have disjoint preimages. Thus only $k = 1$ works.

Complete count for positive k

Value of k	Number of distinct real solutions
$0 < k < 1$	10
$k = 1$	8
$1 < k < 3$	6
$k = 3$	4
$k > 3$	2

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Question 13

How many unordered pairs of distinct integers from $1, 2, \ldots, 30$ have a product that is a perfect square?

A 20

B 24

C 18

D 19

E 16

Worked solution · D

Two positive integers have a square product exactly when the primes appearing to odd powers in their factorisations are the same. Group the integers by that common remaining factor after square factors are removed.

The groups with at least two members are $\{1, 4, 9, 16, 25\}$, $\{2, 8, 18\}$, $\{3, 12, 27\}$, $\{5, 20\}$, $\{6, 24\}$ and $\{7, 28\}$. All other groups are singletons.

The count is $\binom{5}{2} + \binom{3}{2} + \binom{3}{2} + 1 + 1 + 1 = 10 + 3 + 3 + 3 = 19$. Direct enumeration of all $\binom{30}{2} = 435$ pairs gives the same count.

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Question 14

For $0 < x < \pi/2$, the infinite series

$\sin^2 x + \sin^4 x + \sin^6 x + \cdots$

has sum $3\cos^2 x$. What is $\tan^2 x$?

A $\frac{7-\sqrt{13}}{6}$

B $\frac{3}{2}$

C $\frac{\sqrt{13}-1}{2}$

D 3

E $\frac{\sqrt{13}+1}{2}$

Worked solution · C

The common ratio is $\sin^2 x$, which lies strictly between 0 and 1. The sum is $\sin^2 x/(1-\sin^2 x) = \tan^2 x$.

Put $u = \tan^2 x > 0$. Since $\cos^2 x = 1/(1+u)$, the condition becomes $u = 3/(1+u)$. Therefore $u^2 + u - 3 = 0$, and the positive root is $(\sqrt{13}-1)/2$.

The other quadratic root is negative and cannot be a squared tangent. Conversely every positive value of u corresponds to exactly one angle in the specified interval.

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Question 15

A sensor reports height $h(x) = 2x$ for $0 \le x \le 1$, and $h(x) = 6 - x$ for $1 < x \le 2$. It is to be replaced by the constant reading c that minimises $\int_0^2 |h(x) - c|\,dx$. Which option gives both the minimum error and the full interval of minimising values of c?

A $3, \quad 2 \le c \le 4$

B $\dfrac{7}{2}, \quad 2 \le c \le 4$

C $\dfrac{7}{2}, \quad c = 3$

D $4, \quad 1 \le c \le 4$

E $4, \quad c = 3$

Worked solution · B

Pair the points x and $x + 1$ for $0 < x < 1$. Their readings are $2x$ and $5 - x$, with $2x < 5 - x$. The sum of the two absolute errors is at least $5 - 3x$, with equality exactly when $2x \le c \le 5 - x$. Any c in $[2, 4]$ achieves equality throughout. A value outside this interval gives a strict excess on an interval of positive length near $x = 1$. The minimum is $\int_0^1 (5 - 3x)\,dx = 7/2$.

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Question 16

The equation

$3^{2x} - a3^x + 9 = 0$

has distinct real roots u and v satisfying $u^2 + v^2 = 10$. What is a?

A $\frac{28}{3}$

B $\frac{244}{9}$

C 26

D $\frac{82}{3}$

E 28

Worked solution · D

The quadratic in 3^x has roots 3^u and 3^v, whose product is 9. Thus $u + v = 2$. From $u^2 + v^2 = (u+v)^2 - 2uv = 10$, we obtain $uv = -3$.

Consequently u, v are $-1, 3$. The sum of the corresponding positive exponential roots is $a = 3^{-1} + 3^3 = 1/3 + 27 = 82/3$.

These two values are distinct and satisfy the original equation. Vieta's product applies to 3^u and 3^v, not directly to u and v.

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Question 17

Triangle ABC has $AB = AC = 5$ and $BC = 6$, and is inscribed in a circle.

The tangents to the circle at B and C meet at T, on the opposite side of BC from A.

What is AT?

A 20

B $\frac{125}{7}$

C $\frac{50}{7}$

D $\frac{100}{7}$

E $\frac{72}{7}$

Worked solution · D

Choose coordinates $B = (-3, 0)$, $C = (3, 0)$ and $A = (0, 4)$. By symmetry the centre is $(0, h)$. Equating its squared distances from A and B gives $(4 - h)^2 = 9 + h^2$, so $h = 7/8$.

The radius to B has gradient $7/24$, so the tangent at B has gradient $-24/7$. Its intersection with the symmetry axis $x = 0$ is $T = (0, -72/7)$.

Hence $AT = 4 + 72/7 = 100/7$. Check: the radius is $25/8$, and the second tangent gives the same intersection. The location of T satisfies the stipulated side condition.

2026 Predictive Paper · Paper 1

Question 18

Let $f(x) = x^2 - 4x$ and suppose $-4 < a < 0$.

The curves $y = f(|x|)$ and $y = f(x - a)$ enclose a region of area 3.

What is the sum of the squares of all possible values of a?

A 6

B 16

C 12

D 8

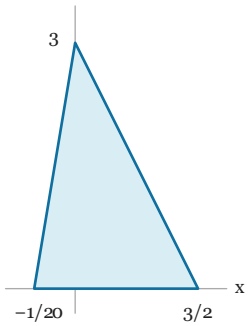
E 10

Worked solution · E

The difference $f(|x|) - f(x - a)$ is $(2a + 8)x - a(a + 4)$ for $x < 0$ and $2ax - a(a + 4)$ for $x \geq 0$. The common x^2 terms cancel.

The intersections are $x = a/2 < 0$ and $x = (a + 4)/2 > 0$. Between them the difference is positive. Its graph is a triangle of base 2 and height $-a(a + 4)$. Thus the enclosed area is $-a(a + 4)$.

Setting this equal to 3 gives $(a + 1)(a + 3) = 0$. Both values lie in $(-4, 0)$, so the required sum is $(-1)^2 + (-3)^2 = 10$. Piecewise integration of the same linear difference gives the identical area.



For a = −1, the vertical difference is triangular. Its area is ½ × 2 × 3 = 3. The a = −3 case is its reflection.

2026 Predictive Paper · Paper 1

Question 19

A monic polynomial P of degree 4 has remainder $2x+1$ when divided by x^2-1, and remainder $x+7$ when divided by x^2-4.

What is the least value of $P(x)+P(-x)$ for real x?

A $\frac{3}{4}$

B 6

C $\frac{3}{2}$

D 3

E $\frac{9}{2}$

Worked solution · C

Write $P(x)=x^4+bx^3+cx^2+dx+e$. Reducing modulo x^2-1 gives $c+e=0$ and $b+d=2$. Reducing modulo x^2-4 gives $4c+e=-9$ and $4b+d=1$.

Thus $c=-3$ and $e=3$. The odd terms disappear in $P(x)+P(-x)=2(x^4-3x^2+3)=2(x^2-3/2)^2+3/2$.

The least value is $3/2$, attained at $x=\pm\sqrt{3/2}$. The unused equations give $b=-1/3$, $d=7/3$, confirming that the specified polynomial exists.

2026 Predictive Paper · Paper 1

Question 20

For which real values of a does

$4^x - 2a\,2^x + a + 3 = 0$

have two distinct real solutions, both strictly between 0 and 2?

A $\frac{1+\sqrt{13}}{2} < a < \frac{19}{7}$

B $\frac{1+\sqrt{13}}{2} \leq a < \frac{19}{7}$

C $1 < a < \frac{19}{7}$

D $\frac{1+\sqrt{13}}{2} < a \leq \frac{19}{7}$

E $\frac{1+\sqrt{13}}{2} < a < 4$

Worked solution · A

Put $t = 2^x$. The required roots must lie in $1 < t < 4$. The quadratic is $q(t) = t^2 - 2at + a + 3$, with vertex at $t = a$.

For two distinct roots in $(1, 4)$, the vertex must lie inside, its value must be negative, and both endpoint values must be positive. These conditions are $1 < a < 4$, $a^2 - a - 3 > 0$, $q(1) = 4 - a > 0$ and $q(4) = 19 - 7a > 0$. They are also sufficient for an upward-opening quadratic.

Combining gives $(1 + \sqrt{13})/2 < a < 19/7$. At the lower boundary the roots merge; at the upper boundary one root is $t = 4$, corresponding to the forbidden endpoint $x = 2$.

2026 Predictive Paper · Paper 2

Question 1

A function f, defined for every real number, has exactly four distinct real zeros.

The equation $f(x + 1/x) = 0$, with $x \neq 0$, has exactly five distinct real solutions.

How many of the zeros of f must lie strictly between -2 and 2?

A 2

B 3

C 0

D 4

E 1

Worked solution · E

A zero r of f contributes solutions of $x^2 - rx + 1 = 0$. It contributes two real solutions if $|r| > 2$, one if $|r| = 2$, and none if $|r| < 2$. No solution is zero, and different values of r have disjoint preimages.

Let h count the outer zeros and b count zeros at the two boundary points. Then $2h + b = 5$, with $0 \leq b \leq 2$. Hence $b = 1$, $h = 2$. Exactly $4 - 1 - 2 = 1$ zero lies inside.

The conditions are feasible: zeros $-3, 0, 2, 4$ produce $2 + 0 + 1 + 2 = 5$ real preimages.

2026 Predictive Paper · Paper 2

Question 2

A student attempts to prove that, for all real a, b,

$$\sqrt{a^2+1}+\sqrt{b^2+1} \geq \sqrt{(a+b)^2+4}$$

The steps are:

I. $(a-b)^2 \geq 0$.

II. Therefore $(a^2+1)(b^2+1) \geq (ab+1)^2$.

III. Therefore $\sqrt{(a^2+1)(b^2+1)} \geq ab+1$.

IV. Therefore $(\sqrt{a^2+1}+\sqrt{b^2+1})^2 \geq (a+b)^2+4$.

V. Taking nonnegative square roots gives the claimed inequality.

Which is the first unjustified step, or is the proof valid?

A V

B III

C II

D IV

E The proof is valid.

Worked solution · E

Step II is equivalent to step I, because $(a^2+1)(b^2+1)-(ab+1)^2=(a-b)^2$. Taking square roots gives a quantity at least $|ab+1|$, and $|ab+1| \geq ab+1$. Thus step III is valid even when $ab+1$ is negative.

On expanding the square in IV and using III, its left side is at least $a^2+b^2+2+2ab+2=(a+b)^2+4$. Both sides in the final comparison are nonnegative, so V is valid.

There is no error. As a check, equality holds when $a=b$. A proof-analysis question need not contain a flaw.

2026 Predictive Paper · Paper 2

Question 3

Let (u_n) be a real sequence indexed by the positive integers. Consider the statement:

"For every real $a \in [1, 2]$, there is a positive integer N such that, for every integer $n \geq N$, either $u_n < a$ or $u_n > 2a$."

Which is its negation?

A There is a real $a \in [1, 2]$ and a positive integer N such that $a \leq u_n \leq 2a$ for every integer $n \geq N$.

B There is a real $a \in [1, 2]$ such that, for every positive integer N, there is an integer $n \geq N$ with $a \leq u_n \leq 2a$.

C There is a real $a \in [1, 2]$ such that, for every positive integer N, there is an integer $n \geq N$ with $a < u_n < 2a$.

D For every real $a \in [1, 2]$ and every positive integer N, there is an integer $n \geq N$ with $a \leq u_n \leq 2a$.

E For every real $a \in [1, 2]$, there is a positive integer N such that $a \leq u_n \leq 2a$ for every integer $n \geq N$.

Worked solution · B

Negation reverses the successive quantifiers: "for every a" becomes "there is an a"; "there is an N" becomes "for every N"; and "for every $n \geq N$" becomes "there is an $n \geq N$".

The negation of $u_n < a$ or $u_n > 2a$ is $u_n \geq a$ and $u_n \leq 2a$. Thus there is some $a \in [1, 2]$ for which every tail of the sequence contains a term in the closed interval $[a, 2a]$.

The endpoints must be included. For example, the sequence alternating between 1 and 0 makes this negation true (take $a = 1$), while each other answer option is false. In particular, no allowed a places either value strictly between a and $2a$.

2026 Predictive Paper · Paper 2

Question 4

Consider the claim:

"Every real polynomial f with $f(0) < 0$, $f(2) > 0$ and $f'(0) > 0$ has exactly one zero in $(0, 2)$."

Which of the following polynomials is a counterexample?

A $f(x) = -4x^3 + 12x^2 - 11x + 3$

B $f(x) = x^3 + x - 1$

C $f(x) = 4x^3 - 12x^2 + 11x - 3$

D $f(x) = x^3 - x - 1$

E $f(x) = 4x^3 - 12x^2 + 8x$

Worked solution · C

The polynomial $4x^3 - 12x^2 + 11x - 3 = (2x - 1)(x - 1)(2x - 3)$ has three distinct zeros $1/2, 1, 3/2$ in $(0, 2)$. Also $f(0) = -3$, $f(2) = 3$ and $f'(0) = 11$, so it satisfies every hypothesis and contradicts the conclusion.

The polynomial $x^3 - x - 1$ fails the derivative hypothesis. The polynomial $x^3 + x - 1$ is strictly increasing, since its derivative is $3x^2 + 1 > 0$, and has exactly one zero in the interval. The polynomial $4x^3 - 12x^2 + 8x$ has $f(0) = f(2) = 0$. The polynomial $-4x^3 + 12x^2 - 11x + 3$ has $f(0) = 3 > 0$.

A valid counterexample must satisfy all assumptions. A positive derivative at one endpoint gives no global monotonicity guarantee.

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Question 5

Let n be a positive integer. Which condition is necessary and sufficient for n^3 to be divisible by 72 while n^2 is not divisible by 72?

A $n = 6k + 3$ for some integer $k \geq 0$.

B n is divisible by 6.

C $n = 12k + 6$ for some integer $k \geq 0$.

D n is divisible by 12.

E $n = 24k + 12$ for some integer $k \geq 0$.

Worked solution · C

Since $72 = 2^3 \cdot 3^2$, divisibility of n^3 requires at least one factor of 2 and at least one factor of 3 in n. Thus n must be divisible by 6.

Its square already has at least two factors of 3. To fail divisibility by 72, it must have fewer than three factors of 2. Therefore n must contain exactly one factor of 2: it is divisible by 6 but not by 12.

This is exactly $n = 12k + 6$ with $k \geq 0$. It is sufficient as well as necessary. For example, $n = 6$ works, whereas $n = 12$ shows that merely requiring divisibility by 6 is insufficient.

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Question 6

Let a, b, c be positive real numbers. Which condition is sufficient but not necessary for a, b, c to be the side lengths of a nondegenerate triangle?

A $1/a, 1/b, 1/c$ are the side lengths of a nondegenerate triangle.

B $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are the side lengths of a nondegenerate triangle.

C a^2, b^2, c^2 are the side lengths of a nondegenerate triangle.

D $a+b, b+c, c+a$ are the side lengths of a nondegenerate triangle.

E The largest of a, b, c is less than the sum of the other two.

Worked solution · C

If a^2, b^2, c^2 form a triangle, then $a^2 < b^2 + c^2 < (b+c)^2$, so $a < b + c$. The other two inequalities follow in the same way. This is sufficient. It is not necessary: $3, 4, 5$ form a nondegenerate triangle, whereas $9, 16, 25$ are degenerate.

The square-root condition is not sufficient: $3, 1, 1$ fail the original triangle inequality, but $\sqrt{3}, 1, 1$ satisfy it. The reciprocal condition is also not sufficient, using the same original triple.

The pairwise-sum condition always holds for positive a, b, c, so cannot guarantee the original triangle. The largest-side condition is both necessary and sufficient, and therefore does not answer the question.

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Question 7

A polynomial f satisfies $0 \le f(x) \le 1$ for $0 \le x \le 1$, and $\int_0^1 f(x)\,dx = 1/2$.

Consider the statements:

I. $\int_0^1 [f(x)]^2\,dx \le 1/2$.

II. $\int_0^1 [f(x)]^2\,dx \ge 1/4$.

III. $\int_0^1 x f(x)\,dx \ge 1/4$.

Which must be true?

A II and III only

B I only

C I and II only

D II only

E I, II and III

Worked solution · C

Since $0 \le f \le 1$, we have $f^2 \le f$ pointwise, proving I. Also $\int_0^1 (f - 1/2)^2\,dx \ge 0$. Expanding and using the given integral gives $\int_0^1 f^2\,dx - 1/4 \ge 0$, proving II.

For III take $f(x) = 1 - x$, which meets all conditions. Then $\int_0^1 x(1-x)\,dx = 1/6 < 1/4$, so III need not hold.

The lower bound in II is attained by $f = 1/2$. No advanced integral inequality is needed: the square of a real expression is nonnegative.

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Question 8

For which real values of a does the polynomial

$f(x) = x^5 - 5ax^3 + 5x$

satisfy $f'(x) > 0$ for every real x?

A $|a| < \frac{2}{3}$

B $a \leq \frac{2}{3}$

C $a > -\frac{2}{3}$

D $a < \frac{2}{3}$

E $|a| \leq \frac{2}{3}$

Worked solution · D

The derivative is $5(x^4 - 3ax^2 + 1)$. Put $t = x^2 \geq 0$ and consider $g(t) = t^2 - 3at + 1$.

If $a \leq 0$, every term except possibly the zero linear term is nonnegative and $g(t) \geq 1$. If $a > 0$, the minimum on $t \geq 0$ is at $t = 3a/2$ and equals $1 - 9a^2/4$. It is strictly positive exactly when $a < 2/3$.

Thus the full answer is $a < 2/3$. At $a = 2/3$, the derivative is $5(x^2 - 1)^2$ and vanishes at $x = \pm 1$, so the boundary is excluded. A negative discriminant is not necessary when t is restricted to nonnegative values.

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Question 9

A continuous function f is defined on the real line. For one fixed positive real number a, it is known that $\int_0^a f(t)\,dt = a^2$.

A student writes:

I. $\int_0^a f(t)\,dt = a^2$.

II. Define $F(x) = \int_0^x f(t)\,dt$; then $F(a) = a^2$.

III. Differentiating this equality with respect to a gives $F'(a) = 2a$.

IV. Therefore $f(a) = 2a$.

You may use the fact that $F'(x) = f(x)$. Which is the first unjustified step, or is the argument valid?

A II

B IV

C The argument is valid.

D III

E I

Worked solution · D

I repeats the hypothesis and II introduces a valid definition. III treats an equality known at one fixed point as though it were an identity valid on an interval. That differentiation is unjustified.

For a concrete counterexample, let $f(x) = 1$ for all real x and take $a = 1$. Then $\int_0^a f = 1 = a^2$, but $f(a) = 1 \neq 2a$.

If instead the original equality were given for every positive a, the differentiation would be valid. The error is in the scope of the hypothesis, not in the stated fact about F'.

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Question 10

Two histograms, A and B, each describe eight real observations. Their class intervals and frequency densities are:

Class interval	$[0, 2)$	$[2, 4)$	$[4, 8]$
Density in A	1	1	1
Density in B	2	1	$1/2$

The median of eight observations is the mean of the fourth and fifth observations in increasing order.

Consider the statements:

I. The mean in A is greater than the mean in B.

II. The median in A is greater than the median in B.

III. The maximum in A is at least the maximum in B.

Which must be true?

A III only

B II only

C I and II only

D I only

E II and III only

Worked solution · B

The frequencies are $2, 2, 4$ for A and $4, 2, 2$ for B. In A, the fourth observation is at least 2 and the fifth at least 4, so the median is at least 3. In B, the fourth is strictly below 2 and the fifth strictly below 4, so the median is strictly below 3. Thus II must hold.

For the other statements, take $A = (0, 0, 2, 2, 4, 4, 4, 4)$ and $B = (3/2, 3/2, 3/2, 3/2, 7/2, 7/2, 8, 8)$. These fit the histograms. Their means are $5/2$ and $29/8$, and their maxima are 4 and 8. Thus I and III both fail.

Exact data values are unknown within their intervals. Nevertheless the strict class endpoints force the median comparison.

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Question 11

A real polynomial f of degree exactly 4 has exactly three distinct stationary points, at $x = a, b, c$ where $a < b < c$.

It satisfies $f(a) = f(c) = 0$ and $f(b) > 0$.

Consider the statements:

I. $b = (a + c)/2$.

II. $f(x) \geq 0$ for every real x.

III. For every real $d > 0$, the equation $f(x) = d$ has exactly four distinct real solutions.

Which must be true?

A II only

B I and III only

C I only

D I, II and III

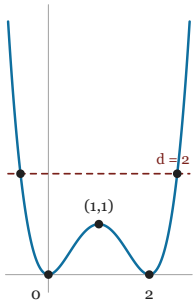
E I and II only

Worked solution · E

Since $f(a) = f'(a) = 0$, the factor $(x - a)^2$ divides f; likewise $(x - c)^2$ divides it. As its degree is exactly four, $f(x) = K(x - a)^2(x - c)^2$ for some nonzero K. The condition $f(b) > 0$ forces $K > 0$, proving II.

The expression is symmetric about $x = (a + c)/2$, where its derivative vanishes. This point is strictly between a and c and must therefore be b, proving I.

Between the two zero minima, the height is at most $f(b)$. For $d > f(b)$ there are just two real intersections, one outside each minimum. At $d = f(b)$ there are three distinct intersections. Thus III is false. The example $f = x^2(x - 2)^2$ confirms these counts.



Example: f(x) = x²(x − 2)². The level d = 2 lies above the middle maximum and meets the curve only twice.

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Question 12

For a real parameter a, consider:

P: For every $x \in [0,1]$, there is a $y \in [0,1]$ such that $(x-y)^2 < a$.

Q: There is a $y \in [0,1]$ such that, for every $x \in [0,1]$, $(x-y)^2 < a$.

For which values of a is P true and Q false?

A $a \geq \frac{1}{4}$

B $a > \frac{1}{4}$

C $0 < a < \frac{1}{4}$

D $0 < a \leq \frac{1}{4}$

E $a \leq 0$

Worked solution · D

P is true exactly when $a > 0$: choose $y = x$. For $a \leq 0$, a nonnegative square can never be strictly smaller than a.

For a fixed y, the largest value of $(x-y)^2$ on $[0,1]$ is $\max(y^2, (1-y)^2)$. Its least possible value over $y \in [0,1]$ is $1/4$, attained at $y = 1/2$.

Because the required comparison is strict and the endpoints are included, Q is true exactly when $a > 1/4$. Therefore P is true and Q false exactly for $0 < a \leq 1/4$.

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Question 13

Suppose $a > 0$, $a \neq 1$, $b > 0$, $c > 0$ and

$\log_a b > \log_a c > 0$

Consider the statements:

I. $(a-1)(b-c) > 0$.

II. $0 < \log_b c < 1$.

III. $b + c > 2$.

Which must be true?

A I, II and III

B II and III only

C II only

D I and II only

E I only

Worked solution · D

If $a > 1$, then $b > c > 1$. Both factors in I are positive. Since $1 < c < b$, the exponent to which b must be raised to give c lies between 0 and 1, proving II in this case.

If $0 < a < 1$, then $0 < b < c < 1$. Both factors in I are negative. As $b^0 = 1$ and $b^1 = b < c < 1$, and powers of this base decrease with the exponent, again $0 < \log_b c < 1$. The base b is never 1, so the logarithm is defined.

III fails, for example with $a = 1/2$, $b = 1/4$, $c = 1/2$, which give logarithms $2 > 1 > 0$ but $b + c = 3/4$. Thus precisely I and II must hold.

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Question 14

A real polynomial f satisfies

$$\int_0^1 [f(x)]^2\,dx = 2\int_0^1 x f(x)\,dx - \frac{1}{3}$$

What is $f(-2)$?

A -1

B It cannot be determined.

C 0

D -2

E 2

Worked solution · D

Since $\int_0^1 x^2\,dx = 1/3$, the given equality is equivalent to $\int_0^1 (f(x) - x)^2\,dx = 0$. The integrand is continuous and nonnegative.

If it were positive at any point of $[0, 1]$, continuity would make its integral positive on a small interval. Hence $f(x) = x$ throughout $[0, 1]$.

The polynomial $f(x) - x$ therefore has infinitely many zeros and must be identically zero. So $f(-2) = -2$. The polynomial assumption is essential for extending the conclusion outside the integration interval.

2026 Predictive Paper · Paper 2

Question 15

For real a, b, c, consider

$$x^2 + y^2 + 2ax + 2by + c = 0$$

Which condition is necessary and sufficient for this equation to describe a nondegenerate circle whose entire circumference lies in the open first quadrant $x > 0$, $y > 0$?

A $a < 0$, $b < 0$, and $\max(a^2, b^2) < c < a^2 + b^2$

B $a < 0$, $b < 0$, and $\max(a^2, b^2) \leq c < a^2 + b^2$

C $a < 0$, $b < 0$, and $0 < c < a^2 + b^2$

D $a < 0$, $b < 0$, and $c > a^2 + b^2$

E $a > 0$, $b > 0$, and $\max(a^2, b^2) < c < a^2 + b^2$

Worked solution · A

Completing the square gives centre $(-a, -b)$ and squared radius $R^2 = a^2 + b^2 - c$. A nondegenerate circle requires $R > 0$. Its circumference lies strictly in the first quadrant exactly when $-a > R$ and $-b > R$.

Thus $a < 0$, $b < 0$, $0 < R^2 < a^2$ and $0 < R^2 < b^2$. Substituting for R^2 gives $c > b^2$, $c > a^2$ and $c < a^2 + b^2$, which combine to the stated condition.

Equality at the lower bound permits tangency to an axis, which the open quadrant excludes. Equality at the upper bound gives a point rather than a nondegenerate circle. A larger value gives no real circle.

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Question 16

A geometric progression has first term $a > 0$ and real common ratio r.

Both infinite series below converge, and

$\sum_{n=1}^{\infty} ar^{n-1} = 2$, while $\sum_{n=1}^{\infty} (ar^{n-1})^2 = 12$.

What is $\sum_{n=1}^{\infty} |ar^{n-1}|$?

A 4

B 6

C 2

D 12

E 8

Worked solution · B

Since $a > 0$ and the series converge, $|r| < 1$. We have $a/(1-r) = 2$ and $a^2/(1-r^2) = 12$. Substituting $a = 2(1-r)$ gives $4(1-r)/(1+r) = 12$.

Thus $r = -1/2$ and $a = 3$. The absolute values form a geometric progression with first term 3 and ratio $1/2$, so their sum is $3/(1-1/2) = 6$.

Check: the original signed sum is $3/(3/2) = 2$ and the sum of squares is $9/(3/4) = 12$. Assuming a positive ratio would rule out the actual solution.

2026 Predictive Paper · Paper 2

Question 17

What is the least integer k such that every selection of k distinct integers from $1, 2, \ldots, 48$ must contain three numbers $a < b < c$ for which a divides b and b divides c?

A 33

B 37

C 36

D 35

E 25

Worked solution · B

Partition the integers into chains obtained by repeatedly doubling an odd starting number. There are 24 chains. The 12 starting with an odd number at most 24 have at least two members; the other 12 are singletons.

A selection without a three-term divisibility chain can contain at most two numbers from each longer doubling chain and at most one from each singleton. It therefore has at most $12(2) + 12(1) = 36$ members. Selecting 37 forces a suitable triple within one chain.

This bound is sharp: the 36 integers $13, 14, \ldots, 48$ contain no such triple. If $a < b < c$ and a divides b and b divides c, then $b \geq 2a$ and $c \geq 2b \geq 4a > 48$. Hence the least k is 37.

2026 Predictive Paper · Paper 2

Question 18

A nonempty finite set S consists of real numbers other than 0 and 1.

For every $x \in S$, both $1 - x$ and $1/x$ also belong to S.

What is the least possible number of elements of S?

A 3

B 5

C 1

D 4

E 2

Worked solution · A

The set $\{-1, 1/2, 2\}$ satisfies both closure conditions, so three elements are possible. A one-element set would require simultaneously $x = 1 - x$ and $x = 1/x$, which is impossible.

Suppose there were two distinct elements x, y. The map $z \mapsto 1 - z$ must permute them. It cannot fix both, since its only fixed point is $1/2$, so it swaps them and $x + y = 1$.

Likewise $z \mapsto 1/z$ must permute the set. Its only allowed real fixed point is -1; it cannot fix both distinct elements, so it must swap them and $xy = 1$. But two real numbers with sum 1 and product 1 would solve $t^2 - t + 1 = 0$, whose discriminant is negative. Thus two are impossible and the minimum is three.

2026 Predictive Paper · Paper 2

Question 19

A continuous function f is defined on the real line and satisfies $f(x+1) = -f(x)$ for every real x.

Consider the statements:

I. $\int_a^{a+2} f(x)\,dx = 0$ for every real a.

II. Every open interval $(a, a+1)$ contains a zero of f.

III. $[f(x+1)]^2 = [f(x)]^2$ for every real x.

Which must be true?

A I and II only

B I, II and III

C II only

D I only

E I and III only

Worked solution · E

The integral over $[a+1, a+2]$ is the negative of that over $[a, a+1]$, by shifting the variable by one and using the given identity. Their sum is zero, proving I. Squaring the identity proves III.

II is false. For example, $f(x) = \sin(\pi x)$ has the required relation but is positive on the open interval $(0, 1)$ and has zeros only at its endpoints there. The identity for this example follows directly from the unit-circle interpretation of sine.

Continuity guarantees a zero in the closed interval $[a, a+1]$, because endpoint values are opposite or both zero. It does not guarantee a zero in the open interval.

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Question 20

For real parameters p, q, which condition is necessary and sufficient for

$$x^4 - px^3 + qx^2 - px + 1 = 0$$

to have four distinct positive real roots?

A $p > 4$ and $2p - 2 < q < \frac{p^2}{4} + 2$

B $p > 0$ and $2p - 2 < q < \frac{p^2}{4} + 2$

C $p > 4$ and $2p - 2 \leq q < \frac{p^2}{4} + 2$

D $p > 4$ and $2p + 2 < q < \frac{p^2}{4} + 2$

E $p > 4$ and $2p - 2 < q \leq \frac{p^2}{4} + 2$

Worked solution · A

Zero is not a root. For positive x, divide by x^2 and put $t = x + 1/x \geq 2$. The equation becomes $t^2 - pt + q - 2 = 0$. Every $t > 2$ gives two distinct positive reciprocal roots; $t = 2$ gives only $x = 1$.

Thus we need two distinct roots of this quadratic strictly greater than 2. Its vertex must lie to the right of 2, its discriminant must be positive, and its value at 2 must be positive. These are $p > 4$, $p^2 - 4q + 8 > 0$ and $q - 2p + 2 > 0$.

Combining yields $p > 4$ and $2p - 2 < q < p^2/4 + 2$. The lower equality introduces $x = 1$ as a repeated root; the upper equality merges the two values of t. A feasible example is $p = 7$, $q = 14$, giving $(x^2 - 3x + 1)(x^2 - 4x + 1)$.